

Carbon and its Compounds

NCERT Class 10 Science Chapter 4 (Hinglish Detailed Notes)
NCERT Class 10 Science Ch 4 | Hinglish Detailed Notes

COLOUR CODE: RED = Core Basics / extra help (Class 7-8-9 wali cheezein jo yaad honi chahiye).

COLOUR CODE: GREEN = Board exam me ye topics BAAR-BAAR aate hain - pakka yaad karo.

Baaki normal black text = main chapter content (detail me, simple bhasha me).

CORE BASICS - PEHLE YE PADHO (CLASS 7-8-9 KA FOUNDATION)

Tune 7-8-9 nahi padhi - koi tension nahi. Ye page is chapter ki base hai.
Ek baar achhe se padh le, fir poora chapter aasaani se samajh aayega.

1) IONIC vs COVALENT BOND (recap):

- IONIC bond = ek atom electron CHHOD-ta, dusra LE leta (metal + non-metal). Opposite charge ke ion attract karte. e.g. NaCl.
- COVALENT bond = dono atom electron SHARE karte (non-metal + non-metal). e.g. H₂O, CH₄. Is chapter me mostly COVALENT bond hi chalega.

2) CARBON KI VALENCY = 4 (sabse important base):

- Carbon ka atomic number 6, electronic config = 2,4 (K shell 2, L shell 4).
- Outer shell me 4 electron -> octet (8) pura karne ko 4 aur chahiye.
- Isliye carbon ki valency 4 hai = TETRAVALENT (4 bond bana sakta).

3) ELEMENT, COMPOUND, MOLECULE:

- ELEMENT = ek hi type ke atom (C, H, O).
- COMPOUND = 2+ element chemically jude fixed ratio me (H₂O, CO₂).
- MOLECULE = 2+ atom jude hue, padaarth ka chhota unit (O₂, CH₄).

4) HYDROCARBON kya hai:

- Sirf CARBON aur HYDROGEN se bana compound = hydrocarbon (CH₄, C₂H₆).
- Petrol, diesel, CNG, LPG sab hydrocarbon hi hain (fuel).

5) SINGLE / DOUBLE / TRIPLE BOND ka matlab:

- SINGLE bond = 1 electron pair share (C-C). DOUBLE = 2 pair (C=C).
- TRIPLE = 3 pair share (C tripletie C). Zyada bond = zyada reactive.

6) STRUCTURAL FORMULA idea:

- Sirf C aur H ki ginti (CH₄) = molecular formula.
- Har bond ko line (-) se dikhana = STRUCTURAL formula (atom kaise jude).

7) COMBUSTION / OXIDATION ka matlab:

- COMBUSTION = oxygen me jalna, heat + light deta (CH₄ + O₂ -> CO₂ + H₂O).
- OXIDATION = oxygen add hona ya hydrogen hatna (alcohol -> acid banna).

8) ACID / ALCOHOL ka intuition:

- ALCOHOL = jisme -OH group ho (sharab/spirit jaisa), e.g. C₂H₅OH.
- ACID = khatta, litmus blue ko red kare; carboxylic acid me -COOH hota.

9) ISOMER ka matlab (intro):

- Same molecular formula par alag STRUCTURE wale compounds = isomers. e.g. C₄H₁₀ ke do roop (seedhi chain aur branched chain).

10) CATALYST kya hai:

- Woh padaarth jo reaction ki SPEED badha de par khud kharch na ho. e.g. hydrogenation me Nickel (Ni) catalyst.

1. COVALENT BONDING - CARBON ELECTRON SHARE KYUN KARTA HAI

Carbon ke outer shell me 4 electron hain. Octet (8) pura karne ke liye:

- 4 electron CHHOD-na padega -> bahut energy chahiye (mushkil).
- 4 electron LE-na padega -> 4 extra electron ko sambhalna mushkil.

Isliye carbon na deta na leta - balki electron SHARE karta = COVALENT BOND.

Electron share karne se dono atom apna octet pura kar lete hain.

1.1 COVALENT BOND ke examples (shared pairs)

- H₂ : H-H (1 pair share = single bond)
- O₂ : O=O (2 pair share = double bond)
- N₂ : N triple N (3 pair share = triple bond)
- H₂O : H-O-H (oxygen 2 H ke saath share karta)
- NH₃ : N ke saath 3 H (nitrogen ke 3 bond)
- CH₄ : C ke 4 bond, 4 H ke saath (methane)

1.2 COVALENT COMPOUNDS ke PROPERTIES (exam favourite list)

- (i) LOW melting point aur boiling point (weak inter-molecular force).
- (ii) Mostly POOR conductor of electricity (koi free ion/electron nahi).
- (iii) Generally paani me kam ghulte, organic solvent me ghulte hain.

Reason: covalent molecule neutral hote, charge-wale ion nahi bante, isliye current conduct nahi karte (ionic compounds ke ulta).

2. VERSATILITY OF CARBON (itne saare compounds kyun?)

Carbon ke LAKHON (millions) compounds hain. 3 main reason:

- (1) CATENATION = carbon ka khud-ke-saath lambi chain/ring banane ka gun.
C-C-C-C... seedhi, branched ya ring me jud sakta (sabse zyada carbon me).
 - (2) TETRAVALENCY = carbon ke 4 bond -> 4 alag atom ke saath jud sakta
(H, O, N, S, halogen) -> bahut variety ke compounds.
 - (3) CHHOTA SIZE = nucleus shared electron ko strongly pakadta -> bahut
STRONG aur stable bond -> compounds tikau (stable) hote hain.
- Isi wajah se carbon "living world" ka base hai (DNA, protein, etc).

3. ALLOTROPES OF CARBON (ek hi element, alag roop)

ALLOTROPE = same element ke alag-alag physical roop (alag structure).

3.1 DIAMOND

- Har carbon, 4 dusre carbon se juda -> rigid 3D structure.
- Sabse HARD natural padaarth, high melting point.
- Electricity conduct NAHI karta (koi free electron nahi).
- Use: cutting/drilling tools, jewellery.

3.2 GRAPHITE

- Har carbon sirf 3 carbon se juda -> layers (parat) banti hain.
 - Layers ek dusre par slide karti -> SMOOTH/slippery (lubricant).
 - Electricity CONDUCT karta (har carbon ka 1 free electron hota).
 - Use: pencil "lead", electrode, lubricant.
- Yaad rakho: diamond hard par insulator; graphite soft par conductor.

3.3 FULLERENE (C60)

- Carbon atoms football (buckyball) jaise gol structure me - C60 famous.
- Naya allotrope, research/nanotech me use.

4. SATURATED vs UNSATURATED HYDROCARBONS

SATURATED = carbon ke beech sirf SINGLE bond (C-C). Inhe ALKANE kehte.

- General formula: C_nH_{2n+2} (e.g. CH_4 , C_2H_6 , C_3H_8).

UNSATURATED = beech me DOUBLE ya TRIPLE bond ho.

- DOUBLE bond (C=C) wale = ALKENE, formula C_nH_{2n} (e.g. C_2H_4 ethene).
- TRIPLE bond wale = ALKYNE, formula C_nH_{2n-2} (e.g. C_2H_2 ethyne).

Unsaturated zyada REACTIVE hote (addition reaction dete hain).

4.1 CHAINS aur RINGS

- STRAIGHT chain: carbon ek line me jude (n-butane).
- BRANCHED chain: beech wale carbon se shaakha nikle (isobutane).
- RING (cyclic): carbon gol ring me jude.
e.g. CYCLOHEXANE (C_6H_{12} ring), BENZENE (C_6H_6 ring with alternate bonds).

5. HOMOLOGOUS SERIES

DEFINITION: Same general formula aur same functional group wale compounds ki series, jisme har agla member pichhle se ek $\text{-CH}_2\text{-}$ unit zyada ho.

CHARACTERISTICS (exam me likhna aata chahiye):

- (i) Sabhi members ka SAME general formula (e.g. alkane $\text{C}_n\text{H}_{2n+2}$).
- (ii) Do lagatar members me $\text{-CH}_2\text{-}$ (yaani 14 u mass) ka farak.
- (iii) Same FUNCTIONAL GROUP $\rightarrow$ multi-julti chemical properties.
- (iv) Molecular mass badhne ke saath physical properties (M.P/B.P) GRADUAL (dheere-dheere) change hoti hain.

Example (Alkane series): $\text{CH}_4 \rightarrow \text{C}_2\text{H}_6 \rightarrow \text{C}_3\text{H}_8 \rightarrow \text{C}_4\text{H}_{10}$ (har baar $+\text{CH}_2$).

6. FUNCTIONAL GROUPS

FUNCTIONAL GROUP = woh atom/group jo compound ko uski khaas property deta hai (hydrogen ya carbon ki jagah lagta hai). Yaad rakho:

- HALOGEN (-Cl, -Br) -> haloalkane (e.g. chloromethane CH_3Cl).
- ALCOHOL (-OH) -> e.g. ethanol $\text{C}_2\text{H}_5\text{OH}$.
- ALDEHYDE (-CHO) -> e.g. ethanal CH_3CHO .
- KETONE (-CO- beech me) -> e.g. propanone CH_3COCH_3 .
- CARBOXYLIC ACID (-COOH) -> e.g. ethanoic acid CH_3COOH .

7. IUPAC NOMENCLATURE (naam rakhne ke rules)

Naam = PREFIX + ROOT (carbon count) + SUFFIX (functional group).

ROOT (carbon ki ginti): 1=Meth, 2=Eth, 3=Prop, 4=But, 5=Pent.

SUFFIX/ending family ke hisaab se:

- Alkane (single bond) -> "-ane" (methane CH₄)
- Alkene (double bond) -> "-ene" (ethene C₂H₄)
- Alkyne (triple bond) -> "-yne" (ethyne C₂H₂)
- Alcohol (-OH) -> "-ol" (methanol CH₃OH)
- Aldehyde (-CHO) -> "-al" (methanal HCHO)
- Carboxylic acid (-COOH) -> "-oic acid" (methanoic acid HCOOH)
- Ketone (-CO-) -> "-one" (propanone CH₃COCH₃)

Example 1-carbon family: methane / methanol / methanal / methanoic acid.

Trick: pehle carbon gino (root), fir functional group dekh ke suffix lagao.

8. CHEMICAL PROPERTIES OF CARBON COMPOUNDS

4 important reactions exam me baar-baar: COMBUSTION, OXIDATION, ADDITION, SUBSTITUTION.

8.1 COMBUSTION (jalna)

Carbon compound O₂ me jal kar CO₂ + H₂O + heat + light deta.



- SATURATED hydrocarbon -> saaf NEELI (blue) flame (poora oxygen mile to).
- UNSATURATED / kam oxygen -> kaali dhuaan wali PEELI (yellow) sooty flame.

Gas stove ki blue flame = clean burning; bartan kaala ho to oxygen kam.

8.2 OXIDATION

Alcohol ko OXIDISE karne par CARBOXYLIC ACID banta:



- OXIDISING AGENT = alkaline KMnO₄ ya acidified K₂Cr₂O₇.
- Ye agent oxygen dete hain, isliye "oxidising agent" kehte.

8.3 ADDITION REACTION (sirf UNSATURATED par)

Unsaturated hydrocarbon + H₂ --[Ni/Pd catalyst]--> saturated hydrocarbon.

(HYDROGENATION) e.g. ethene + H₂ -> ethane.

USE: vegetable oil (unsaturated, liquid) + H₂ (Ni) -> VANASPATI GHEE (saturated, solid).

Isliye ise "hardening of oils" kehte hain - oil thos ban jaata.

8.4 SUBSTITUTION REACTION (sirf SATURATED par)

Saturated hydrocarbon ke H ki jagah dusra atom aa jaaye (sunlight me):



- Ek-ek karke H ki jagah Cl lagta jaata (chlorination).

9. ETHANOL (C₂H₅OH)

PROPERTIES: rang-heen liquid, room temp par liquid, paani me poori tarah ghulta.
Ye hi "alcohol" hai jo sharab (alcoholic drinks) me hota.

9.1 SODIUM ke saath reaction:

$2\text{C}_2\text{H}_5\text{OH} + 2\text{Na} \rightarrow 2\text{C}_2\text{H}_5\text{ONa} + \text{H}_2$ (gas)
(sodium ethoxide + hydrogen gas nikalti) - alcohol ka test.

9.2 DEHYDRATION (paani hatna) - ethene banana:

$\text{C}_2\text{H}_5\text{OH} \xrightarrow{\text{conc. H}_2\text{SO}_4, 443\text{K}} \text{C}_2\text{H}_4$ (ethene) + H_2O
conc. H_2SO_4 yahan dehydrating agent hai.

USES: alcoholic drinks, solvent, medicines/tincture, fuel.

DENATURED ALCOHOL = ethanol me zaharili cheez (methanol) mila dete taaki
peene laayak na rahe (industry use ke liye, tax bachane).

HARMFUL EFFECTS: jyada peena liver kharab karta, judgement/health bigadta,
pure methanol to andhapan/maut tak kar sakta - bahut khatarnaak.

10. ETHANOIC ACID (CH₃COOH) - ACETIC ACID

Sirka (VINEGAR) = ethanoic acid ka 5-8% paani wala ghol.

GLACIAL ACETIC ACID = pure (almost 100%) ethanoic acid; thandi me jam jaata (ice jaisa).

10.1 NaOH (base) ke saath - neutralisation:



10.2 CARBONATE / BICARBONATE ke saath (CO₂ nikalti - bubbles):



(Ye fizz/CO₂ wala test acid pehchanne me kaam aata.)

10.3 ESTERIFICATION (sweet smell - exam favourite):



(ester = ethyl ethanoate; meetha/fruity smell - perfume aur flavour me use.)

10.4 SAPONIFICATION (ester + base → soap):



Ye reaction soap banane ka base hai.

11. SOAPS & DETERGENTS

SOAP MOLECULE ke 2 part hote:

- HYDROPHILIC head ($-\text{COONa}$) = paani-pasand (ionic, water me ghulta).
- HYDROPHOBIC tail (lambe carbon chain) = paani-se-door, oil/grease pasand.

11.1 MICELLE banna:

Paani me soap ke molecule cluster banate $\rightarrow$ tails andar (oil ki taraf), heads bahar (paani ki taraf). Is gol cluster ko MICELLE kehte.

11.2 CLEANSING ACTION (kaise saaf karta):

Tail kapde ki GANDAGI/oil ko pakadta, head paani ki taraf rehta. Micelle me phans ke gandagi paani me ghul/uthkar nikal jaati $\rightarrow$ kapda saaf.

11.3 HARD WATER ka problem (soap):

Hard water me Ca/Mg salts hote. Soap inse react kar ke insoluble SCUM (chipchipe jhaag-rahit kachra) bana deta $\rightarrow$ jhaag nahi banta, saboon barbaad.

11.4 DETERGENTS ka faida:

Detergent hard water me bhi kaam karte (Ca/Mg ke saath scum nahi bante), isliye hard water me bhi achha jhaag/safai dete.

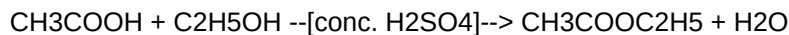
Soap = natural (vegetable/animal fat se); Detergent = synthetic.

12. SOLVED EXAMPLES (HARDEST -> EASIEST)

Ye type ke questions exam me pakka aate - steps khud likhke practice karo.

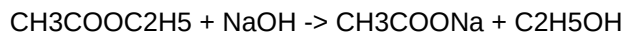
EXAMPLE 1 (Hardest): Esterification aur Saponification ki poori equation likho.

ESTERIFICATION (acid + alcohol -> ester):



(ethyl ethanoate ester banta - meetha smell.)

SAPONIFICATION (ester ka base se ulta-toot-na -> soap):



(CH₃COONa = soap/sodium salt, alcohol wapas milta.)

EXAMPLE 2: C₄H₁₀ ke structural isomers banao/pehchaano.

Molecular formula same (C₄H₁₀) par 2 structure:

(a) n-BUTANE : C-C-C-C (seedhi straight chain).

(b) ISO-BUTANE: beech wale carbon se ek CH₃ branch (branched chain).

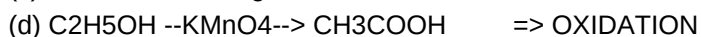
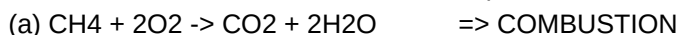
Dono isomers hain - same formula, alag structure, alag properties.

EXAMPLE 3: Ethanol aur Ethanoic acid me farak kaise pehchaane (tests)?

(i) LITMUS test: ethanoic acid blue litmus ko RED kar deta; ethanol nahi (neutral).

(ii) NaHCO₃ test: ethanoic acid + NaHCO₃ -> CO₂ ki FIZZ (bubbles); ethanol me koi fizz nahi. (Isse pakka pehchaan ho jaata.)

EXAMPLE 4: In reactions ka TYPE batao (combustion/oxidation/addition/substitution):



EXAMPLE 5: Vegetable oil ka hydrogenation - equation + use.

Vegetable oil (unsaturated) + H₂ --[Ni catalyst]--> Vanaspati ghee (saturated).

Use: liquid oil ko thos ghee me badalne ke liye (hardening of oils).

EXAMPLE 6: Carbon itne saare compounds kyun banata hai?

(i) CATENATION - carbon khud se lambi chain/ring banata.

(ii) TETRAVALENCY - 4 bond, alag-alag atom ke saath jud sakta.

(iii) Chhota size -> strong stable bond. Inhi se lakhon compounds.

EXAMPLE 7 (Easiest): Methane ke baad alkane series ka agla member + general formula?

General formula alkane = C_nH_{2n+2}. CH₄ ke baad agla = C₂H₆ (ethane).

(Har agla member +CH₂ zyada hota - homologous series.)

13. QUICK Q&A - PADHNE KE BAAD KHUD CHECK KARO

Pehle khud answer socho, fir niche milao. Ye board-style questions hain.

Q1. Carbon ionic bond kyun nahi banata, covalent kyun?

A1. 4 electron dena/lena bahut mushkil (energy zyada), isliye carbon electron SHARE karta (covalent) aur octet pura karta.

Q2. Catenation kya hai?

A2. Carbon ka khud-ke-saath lambi chain/ring/branch banane ka gun - isi se bahut saare compounds bante.

Q3. Saturated aur unsaturated hydrocarbon me farak?

A3. Saturated = sirf single bond (alkane, C_nH_{2n+2}). Unsaturated = double/triple bond (alkene C_nH_{2n} / alkyne C_nH_{2n-2}), zyada reactive.

Q4. Homologous series ki koi 2 characteristics likho.

A4. (i) Same general formula + same functional group. (ii) Do members me $-CH_2-$ (14 u) ka farak aur gradual property change.

Q5. Saaf neeli flame aur peeli sooty flame me kya batati hai?

A5. Neeli flame = poora oxygen, clean combustion (saturated). Peeli/sooty = oxygen kam ya unsaturated - kaala dhua/carbon nikalta.

Q6. Ethanol ko sodium ke saath react karaao - product?

A6. $2C_2H_5OH + 2Na \rightarrow 2C_2H_5ONa + H_2$. Hydrogen gas nikalti (sodium ethoxide banta).

Q7. Esterification reaction me catalyst aur khaas pehchaan?

A7. Catalyst = conc. H_2SO_4 . Pehchaan = ester ka meetha/fruity smell (perfume me use).

Q8. Soap hard water me theek se kaam kyun nahi karta?

A8. Hard water ke Ca/Mg salts soap se react kar insoluble SCUM banate $\rightarrow$ jhaag nahi banta. Detergent hard water me bhi chalta.

Q9. Micelle kya hai?

A9. Paani me soap molecules ka gol cluster - tail andar (oil ki taraf), head bahar (paani ki taraf). Yahi gandagi ko ghol kar nikalta.

Q10. Glacial acetic acid kya hai?

A10. Pure (almost 100%) ethanoic acid - thandi me ice jaisa jam jaata isliye "glacial" kehte hain.

EXAM STRATEGY (1 minute revision):

- Covalent bonding + catenation/tetravalency reasons - 3-3 line ratlo.
- 4 reactions (combustion, oxidation, addition, substitution) + 1-1 equation.
- Ethanol (Na, dehydration) aur Ethanoic acid (NaOH, carbonate, ester) ki reactions.
- Homologous series + functional groups + IUPAC suffix table.
- Soap micelle + cleansing action + hard water ka problem.
- CORE BASICS page (valency 4, covalent vs ionic, isomer) bhool jaao to

wapas pehle padho - base strong rakho.

Generated by Kiro for Krishna1k - All the best!